Question			Answer	Marks	AOs	Guidance	
6	(a)		$v = t^3 - 5t^2 = 0$ $t^2(t-5) = 0$ So at rest when $t = 0$ or 5 s	**M1** **A1** **[2]**	**1.1a** **1.1**	**DR** Attempt to solve $v = 0$ Both roots and no others	Allow M1 for at least one root found by substitution
6	(b)		There is a change of direction when $t = 5$ $\int_0^5 \left(t^3 - 5t^2\right) \mathrm{d}t = \left[\frac{t^4}{4} - \frac{5t^3}{3}\right]_0^5 = -\frac{625}{12}$ $\int_5^6 \left(t^3 - 5t^2\right) \mathrm{d}t = \frac{193}{12}$ Total distance $\frac{625+193}{12} = \frac{409}{6} = 68.2$ m	**M1** **M1** **A1** **[3]**	**3.4** **3.4** **1.1**	**DR** Considering signed areas either side of $t = 5$ s. Algebraic integration seen attempted Correct to at least 2 sf	SPECIAL CASE $\int_0^6 \left(t^3 - 5t^2\right) \mathrm{d}t$ $= \left[\frac{t^4}{4} - \frac{5t^3}{3}\right]_0^6 = -36$ **SCM1** for algebraic integration seen attempted **SCA1** for −36 m seen or distance 36 given